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Realistic model of a solid substrate fermentation packed-bed pilot bioreactor

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Abstract

For any given large-scale solid substrate fermentation (SSF) bioreactor, to assess how well a control system will work in practice requires the most realistic model possible. This model needs to account fully for complicated dynamic reactor behaviour and, in addition, has to include a specific noise model that is capable of reproducing the disturbances observed in SSF bioreactor measurements. In this work, noisy data collected historically from SSF pilot scale fermentations was used to develop such a model. Applying standard signal processing techniques, each measured variable was separated into deterministic and noise signals. Deterministic signals were used to calibrate a previously developed phenomenological model of the bioreactor. Noise signals were used to construct a realistic noise model for each measured variable in turn. Finally, the two models were combined to attain simulations that compared well with real measurements. This integrated model will provide realistic simulations that will prove useful in the design of effective control systems for intermittently mixed SSF bioreactors.
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1. Introduction

Mankind has been cultivating microorganisms on solid substrates for centuries. Nevertheless, nowadays industry makes use of many more submerged fermentation (SMF) applications than solid substrate fermentation (SSF) processes. At first sight this is difficult to comprehend since SSF presents a number of economic, biotechnological, and environmental advantages over SMF [1,2]:

- higher volumetric productivities, thus requiring smaller bioreactors;
- fermentation time are usually shorter;
- reduced contamination by bacteria and yeasts;
- reduced metabolite degradation during the fermentation process;
- high concentration of metabolites in the final solid substrate renders further processing needless;
- certain metabolites are expressed in SSF only;
- SSF is better adapted to mixed-microbial cultures;
- SSF fermentation products (enzymes, spores and metabolites) generally have a longer shelf life;
- low water consumption of SSF processes, therefore reducing liquid effluent treatment costs;
- SSF can use agricultural wastes as substrates.

Many authors have reported remarkable results in terms of yield or quality when cultivating microorganisms on solid substrates at laboratory scale (a few grams in glass columns). However, when these processes are reproduced at larger scales (100 kg or more) results are mostly disappointing. Clearly scaling-up SSF processes is difficult and unreliable [1–5]. This can be attributed to the radically different growing conditions microorganisms find in a large scale reactor. At lab scale, conditions in glass columns such as temperature, water activity, and pH can easily be maintained homogeneous and constant (space and time, respectively) at optimum levels throughout the fermentation processes [3–5]. In contrast, bed heterogeneity in large scale SSF bioreactors is inevitable, as simulations with a distributed parameter model [6] and experiments with a pilot scale SSF bioreactor [7] have shown. This is due to the low heat and mass transfer rates characteristic of the solid porous bed and that stirring is limited to avoid damaging the microorganisms [1,3–5,8]. Effective control strategies have been shown to be useful to reduce time and space variability in growing conditions in large scale SSF bioreactors [9,10]. However, process complexity and the lack of reliable and affordable

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instrumentation make designing control strategies difficult. Moreover, even variables that can often be measured reliably, for instance temperature, in large scale SSF bioreactors are extremely noisy due to bed channelling, gas occlusion inside the porous bed, periodic bed stirring and electromagnetic interference [11–13]. Finally, real fermentation times are long (several days), with the consequence that testing and tuning new control strategies is expensive and time consuming. Computer simulation, on the other hand, is a good option for testing new automatic control designs, since even hundreds of simulation runs are fast and relatively inexpensive. Nevertheless, only simulations with realistic models provide credible results.

Both lumped parameter models [5,14–17] and distributed parameter models [6,10,18] have been proposed for SSF processes. Most of these models have, however, only been tested at laboratory scale [3,19–21] given the difficulties found in the instrumentation and control of large scale bioreactors [22]. Lekanda and Pérez-Correa [23] experimentally tested a model for a pilot scale SSF bioreactor (200 kg). Their model includes the mass and energy balances developed in [5] and the kinetic model developed in [15]. The main limitation of a purely phenomenological approach is that as it overlooks measurement noise, simulated variables do not represent the complex behaviour observed in real measurements. A further point is that their model is not fully predictive, since outlet air temperature is considered an input variable.

Here we extend the Lekanda and Pérez-Correa model with a heat transfer model, to predict outlet air temperature, and additionally provide a noise model, so that simulated variables reproduce the complex behaviour observed in real measurements. The complete model comprises a phenomenological deterministic submodel and a noise submodel. The submodels were calibrated with experimental data obtained from a pilot scale (200 kg) SSF bioreactor in which the filamentous fungi *Gibberella fujikuroi* was grown on wheat bran and starch under nitrogen limitation.

In Section 2 we briefly describe both submodels, the parameter estimation methods used for calibration and the signal processing tools applied. Section 3 presents estimated parameters, a sensitivity analysis and comparisons of simulations with real data are made.

2. Materials and methods

2.1. Process description

Experimental data was drawn from several fermentation runs performed in a pilot scale SSF bioreactor described in Fig. 1. The figure also shows auxiliary equipment, the variables measured and where each is in the experimental system. In the bioreactor the filamentous fungi *G. fujikuroi* was cultivated on wheat bran and starch under nitrogen limitation to produce gibberellic acid, a growth hormone.

Certain indicated variables were measured on-line with a sampling time of 20 s. In addition, other important process variables were measured manually: bed-water content (every 4 h), water seepage from the bottom of the reactor (every 8 h), and total dry mass (at the beginning and at the end of the fermentation). The bioreactor was stirred occasionally to reduce temperature gradients and to homogenize bed moisture when fresh water was added. Bed temperature was maintained at 28 °C by evaporative cooling, i.e., by manipulating inlet air relative humidity to control water evaporation. This cooling strategy requires periodic addition of fresh water to replenish evaporation losses. Further details regarding this process can be found in [5,23,24].

We possess historical data from nine fermentation runs of *G. fujikuroi* carried out in the 200 kg pilot scale SSF bioreactor described above [23].

2.2. Phenomenological model

This work considers the cultivation of *G. fujikuroi* on wheat bran and starch in an aseptic pilot scale bioreactor. Details of a similar model can be found in [23]. Here, we only provide a brief description and highlight the differences with the previous model. The current model comprises a kinetic submodel [15] and a macroscopic submodel that includes mass and energy balances [5], resulting in a set of algebraic and ordinary differential equations.

2.2.1. Kinetics

The main assumptions regarding the kinetic relationships are: homogeneous system; oxygen transfer resistance is negligible; available nitrogen is the limiting substrate; model parameters remain constant throughout cultivation

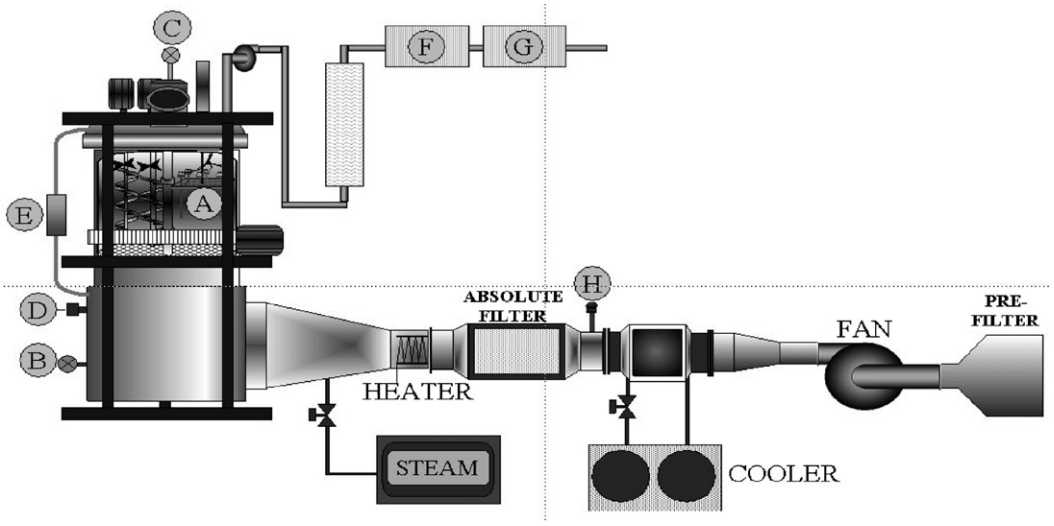


Fig. 1. SSF pilot bioreactor instrumentation. (A) Six thermocouples measure solid-bed axial and radial temperature distribution; (B) thermocouple measures inlet air temperature; (C) thermocouple measures outlet air temperature; (D) inlet air relative humidity measurement; (E) solid bed pressure drop measurement; (F) CO_2 measurement; (G) outlet air O_2 measurement; (H) inlet air velocity measurement.

and temperature and water activity do not affect the kinetic parameters [1,3–5,15,25–29]; all variables are expressed on a dry mass basis.

Total biomass (X_{tot}) includes active (X) and inactive fungi:

$$\frac{dX_{tot}}{dt} = \mu X \ (kg_X/h\,kg_{d.b.}) \quad (1)$$

$$\frac{dX}{dt} = \mu X - \delta X \ (kg_X/h\,kg_{d.b.}) \quad (2)$$

Consumption of available nitrogen can be expressed by

$$\frac{dN_I}{dt} = \phi k - \mu\left(\frac{X}{Y_{x/N_I}}\right) \ (kg_N/h\,kg_{d.b.}) \quad (3)$$

Evolution of respiratory gases, carbon dioxide production and oxygen consumption, involves two terms, growth and maintenance:

$$\frac{dCO_2}{dt} = \mu\frac{X}{Y_{x/CO_2}} + m_{CO_2}X \ (kg_{CO_2}/h\,kg_{d.b.}) \quad (4)$$

$$\frac{dO_2}{dt} = \mu\frac{X}{Y_{X/O_2}} + m_{O_2}X \ (kg_{O_2}/h\,kg_{d.b.}) \quad (5)$$

Dry mass degradation rate is linearly related to total CO_2 production rate (CPR):

$$\frac{dM_s}{dt} = -k_g CPR \ (kg_{d.b.}/h) \quad (6)$$

The rate of metabolic production of water is also proportional to the specific CPR:

$$R_W = \frac{(k_W CPR)}{M_s} \ (kg_W/h\,kg_{d.b.}) \quad (7)$$

Monod's rate expression is used to model the effect of the limiting nutrient (available nitrogen) on the specific growth rate (μ):

$$\mu = \frac{\mu_M N_I}{N_I + k_N} \ (1/h) \quad (8)$$

2.2.2. Water and energy macroscopic balances

We assume the outlet air is saturated, gas phase accumulation is insignificant and that physical properties remain constant during cultivation. Water and temperature gradients in the solid bed are neglected. Reactor walls are considered part of the fermenting bed in the energy balance. The macroscopic model consists of an overall water balance and an energy balance, as follows.

The bed water balance considers metabolic production (R_W), direct addition of fresh water (F_W), evaporation (E), seepage (W_p), and solid bed degradation (the last term in Eq. (9)):

$$\frac{dX_W}{dt} = R_W - E + \frac{F_W - W_p}{M_s} - \frac{X_W}{M_s}\frac{dM_s}{dt} \ (kg_W/h\,kg_{d.b.}) \quad (9)$$

Evaporation can be estimated directly from the inlet and outlet gas conditions:

$$E = \frac{G(Y_{go} - Y_{gi})}{M_s} \ (kg_W/h\,kg_{d.b.}) \quad (10)$$

where Y_{gi} and Y_{go} are the absolute humidities of the inlet and outlet air streams, respectively. These were calculated from classical thermodynamic relationships [30].

The energy balance also contains several terms: metabolic heat generation (ΔHr), air cooling (Q_g), comprising forced convection and water evaporation, wall cooling (Q_{wall}), involving natural convection and radiation, as well as other minor terms associated with the enthalpy of adding fresh water (Q_{F_W}), and the enthalpy of respiratory gases (Q_{O_2} and Q_{CO_2}):

$$C_s k_{exp}\frac{dT_b}{dt} = Q_{F_W} + \Delta H_r + Q_{O_2} - Q_{CO_2} - Q_g - Q_{wall} \ (J/h) \quad (11)$$

The total heat capacity (C_s), in the previous term for energy accumulation, includes the bioreactor wall, given their high thermal conductivity and high thermal capacity:

$$C_s = M_s C_{P,S} + M_s X_W C_{P,W} + C_{P,wall}M_{wall} \ (J/K) \quad (12)$$

Additionally, the parameter k_{exp} accounts for the difference between the average bed temperature and the average of the six measured temperatures in the bed.

Metabolic heat generation is proportional to CPR:

$$\Delta H_r = -Y_{q/Co_2}CPR \ (J/h) \quad (13)$$

The gas cooling term takes 0 °C as the reference temperature:

$$Q_g = G[C_{P,a}(T_{go} - T_{gi}) + C_{P,v}(Y_{go}T_{go} - Y_{gi}T_{gi}) + (\lambda_W - C_{P,v}T_{bprom}) \times (Y_{go} - Y_{gi})] \ (J/h) \quad (14)$$

Here, instead of assuming that the outlet gas temperature (T_{go}) is known, like in [23], we estimated outlet gas temperature with a simple heat transfer model:

$$Q_g = Ah(T_b - \bar{T}_g) \ (J/h), \qquad \bar{T}_g = \frac{T_{gi} + T_{go}}{2} \ (K) \quad (15)$$

Wall cooling includes radiation and natural convection:

$$Q_{wall} = A[(h_{air}(T_{wall} - T_{surr}) + \sigma\varepsilon(T_{wall} + 273.15)^4 - (T_{surr} + 273.15)^4)] \ (J/h) \quad (16)$$

The heat transfer coefficient (h_{air}) was obtained from the Churchill and Chu correlation [31].

Minor terms associated with water addition and respiratory gases are given by

$$Q_{F_W} = F_W C_{P,W}(T_W - T_b) \ (J/h) \quad (17)$$

$$Q_{CO_2} = CPR C_{P,CO_2}(T_{go} - T_b) \ (J/h) \quad (18)$$

$$Q_{O_2} = OCR C_{P,O_2}(T_{gi} - T_b) \ (J/h) \quad (19)$$

2.2.3. Model calibration

All of the measurements referred to above (Section 2.1) were used to calibrate the model using weighted least squares. In the cost function, the sum of squared errors (SSE) for each measured variable was computed using:

$$SSE(i) = \sum_{i\in l}\left[\frac{X(i) - Y(i)}{Y(i)}\right]^2 \quad (20)$$

where $X(i)$ is the simulated value, $Y(i)$ the measured value at time interval i and l is the number of measured values for the corresponding variable.

Each variable in the cost function needs to be weighted to reflect different optimization priorities for temperature, bed water content and total mass and all of the other variables, and the different number of samples available for each. The sums of the respective squared errors (Eq. (20)) were, therefore, weighted independently. For instance, since we only had the final measurement for total mass (a variable that has to be simulated accurately) its corresponding squared error was multiplied by 1000. In addition, since we measured temperature at six locations inside the solid bed, the least we can expect from our model is that the simulated average bed temperature never breaches the maximum or minimum measured value at any given time. Hence, the cost function (Eq. (20)) penalizes the simulated bed temperature when it exceeds maximum measured bed temperature or when it falls below the minimum measured bed temperature.

Coded in Matlab, optimal model parameters were found using the Optimization Toolbox '*fminsearch*' function.

2.2.4. Sensitivity analysis

To get non-linear models to converge the calibration routine requires initial model parameter guesses to be suitably close to the optimal values. This is difficult to achieve with complex, highly non-linear models containing numerous parameters, which all have an impact on multiple output variables. To help us establish a reasonable starting point for model calibration, we therefore defined $En(i)$ as a normalized error for each measured variable:

$$En(i) = \frac{(\sqrt{SSE(i)^*} - \sqrt{SSE(i)})}{l} \times 100 \quad (21)$$

to study the impact of the model parameters on measured variables. $SSE(i)^*$ and $SSE(i)$ are the sum of squared errors for measured variable i, as defined in Eq. (20), obtained with the initial model parameter and with the disturbed

parameter, respectively. If En(i) is positive, the disturbed parameter improves the calibration for variable i, while negative values of En(i) indicate poorer calibration. Each model parameter was disturbed independently at set proportions (−40%, −25%, −5%, +5%, +25%, and +40%) and the simulation repeated for each.

2.3. Signal processing

Measuring devices are usually connected to data acquisition systems in industrial scale processes so values can be processed online and saved for further offline analysis. Primary signals streams from measuring devices (thermocouples, Pt100, pH electrodes, paramagnetic cells, etc.) are not directly interpretable, and have to be processed (amplified, normalized and sampled). Specific circuitry employed in processing this data can add high frequency noise to the measured signal, however. In SSF processes, motor activation, bed mixing, air channelling and occluded gas may also add high frequency noise to measurements. In addition, electromagnetic interference (EMI) can generate outliers (gross errors) or impede data acquisition causing data loss. It is worth stressing that all of these disturbances also affect variables that can usually be measured reliably, since the effects are independent of the quality of the measuring device. Biased models or poor control performance will result from direct use of contaminated data in the process model or process control. Consequently, measured signals should be freed of high frequency noise, missing data and outliers prior to calibrating a process model or computing a control action. The specific methodology we used to treat raw data from an SSF pilot bioreactor is described next. The methodology includes assigning missing data values, shaving outliers and filtering high frequency noise. A realistic disturbance model for noise and outliers based on statistical analysis was formulated and constructed.

2.3.1. Missing data

Missing data points are measurements that were either not recorded by the data acquisition system or were recorded but return constant values for time durations of up to 1 min and therefore lack physical meaning. Outliers are not considered to be missing data.

Several authors have dealt with the optimal assignment of missing values [32–34]. Most proposed methods employ the maximum expectation (EM) algorithm [35], which however is difficult to tune and code when many, and especially contiguous missing values are entailed. Instead, we applied a novel and simple heuristic algorithm that we have used successfully to complete time series involving many missing values [11].

Consider a time series of recorded values (A, B, C, A', B' and C') with missing values in between (x and y), as shown in Fig. 2.

The method we propose assigns the most probable value of the missing point using the values and estimated variances of the equidistant neighbouring points on either side of the missing value. To estimate the value of x, the algorithm considers values B and A'. In addition, the variances of B, σ_B, and A', σ'_A, are in turn estimated from their neighbouring points. Thus, the assigned value for x can be estimated as a weighted average [36]:

$$x = \frac{B\sigma_{A'}^2 + A'\sigma_B^2}{\sigma_{A'}^2 + \sigma_B^2} \tag{22}$$

Once the value of x has been assigned, the algorithm proceeds to the next missing point (y in Fig. 2). The algorithm concludes when all missing points have been given a value.

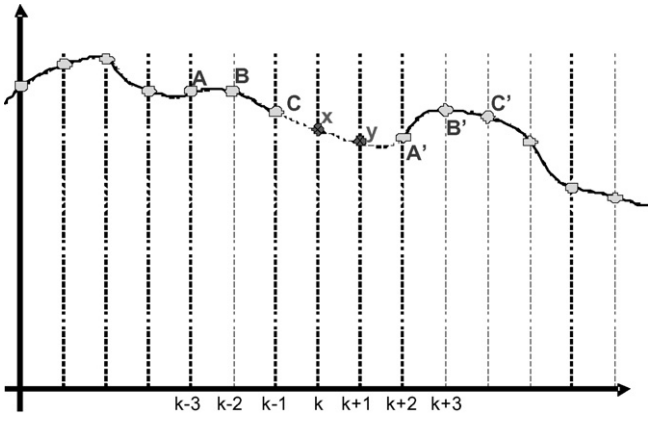


Fig. 2. Time series with missing values.

2.3.2. Outliers

To deal with outliers, it is first necessary to detect when these anomalous isolated values, which appear in the time series without apparent cause, occur. Abrupt changes in the measured signal can be detected if upper and lower thresholds are defined over a search area of the sample. Since both the average value and the standard deviation are sensitive to outliers we used the median and the median absolute deviation (MAD) to compute thresholds over a moving window. Pearson [37] suggests the following formula to compute the MAD value:

$$\text{MAD} = 1.483 \cdot \underset{j=1,\ldots,n}{\text{median}}\{|x_j - \underset{j=1,\ldots,n}{\text{median}}(x_j)|\} \tag{23}$$

where 1.483 is a correction factor to force results to be consistent with a normal distribution. A measurement is considered an outlier when its absolute distance from the median is more than three times the MAD value defined in Eq. (23). Once the outlier is detected, its value is replaced with the median for that point. This, a widely applied algorithm in industrial processes, is known as Hampel identification and is described in further detail in [37,38].

2.3.3. High frequency noise filtering

For filtering high frequency noise non-causal filters are preferred since they do not incur the delay causal filters introduce. Non-causal filters can only be used when historic data is available, which was our case. The Savitsky–Golay polynomial smoothing algorithm [39] is a non-causal filter commonly used in chemical processes and chemometrics. In the algorithm, each value of the time series is estimated as a polynomial approximation defined over a moving window, whose central point is the estimated (filtered) value. The polynomial is fitted using least squares, resulting in optimal filtering when the sampling time is constant. Polynomial order and moving window length are the only user settable tuning parameters.

2.4. Noise modelling

Noise removed using the filtering procedure (described above) provided the data to develop a noise model for each signal measured. We now go on to explain how we used historic data to develop these models.

2.4.1. Outliers

Without exception, every measured signal included outliers at three different magnitudes: high, medium and low. Each group is distributed differently in time and magnitude. At first glance it may seem pointless to distinguish among high, medium and low outliers, but our aim was to reproduce the time distribution of the entire set of outliers in our data and since each outlier group appeared at a different frequency, we had to model each group independently. To assign outliers automatically to one of these three levels, we arbitrarily defined two threshold values (a and b) for each measured variable, so that values above a were considered high outliers, those found between a and b were declared medium outliers and those below b were considered low outliers. Values of high and medium outliers were approximately constant and outliers appeared evenly distributed over the timeframe. Outliers are therefore simply characterized by two parameters: their average values (μ_1 and μ_2) and their frequency (n_1 and n_2). Low outliers, in turn, are distributed normally in magnitude and exponentially in time; hence low outliers are characterized by three parameters, the mean magnitude (μ_2), the standard deviation (σ) and the mean of the time exponential distribution (μ_t). We analyzed data from the nine fermentation runs to estimate parameters for each measured signal. Fig. 3 shows a typical inlet air temperature signal database record. For this particular

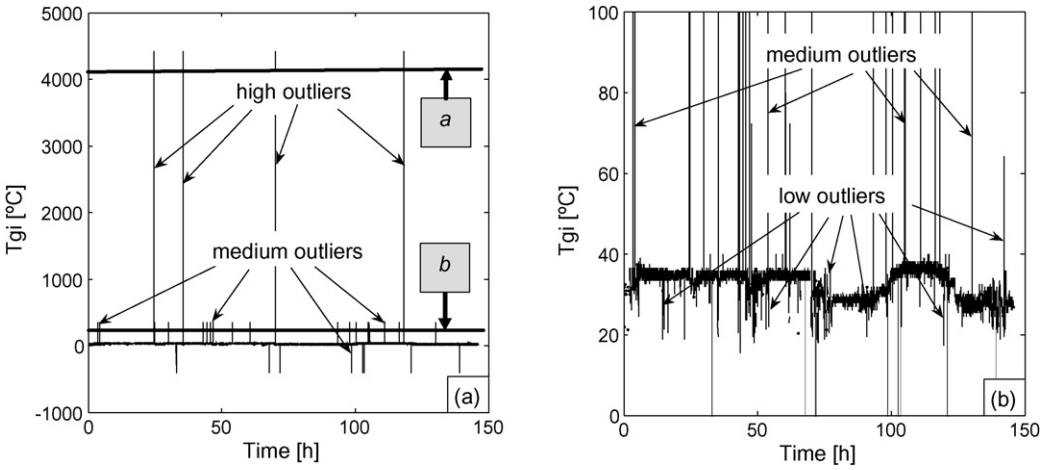


Fig. 3. Selection of high and medium outliers from inlet air temperature of a typical fermentation run. (a) High and medium outliers; (b) medium and low outliers.

Table 1
Model parameter sensitivity on simulated values

	δ	k	Y_{x/CO_2}	Y_{x/O_2}	m_{CO_2}	m_{O_2}	k_P	μ_M	k_N	k_g	k_W	Y_{q/CO_2}
Bed water content		+	+		+			+	+	++	++	++
Bed temperature band between T_{max} and T_{min}	+	++	+	+	+	+		++	++	+		++
Average bed temperature	++	++	++		++			++	+	+		++
Outlet air temperature	++	++	++		++			+	+	+	+	++
Outlet CO_2 concentration	++	++	++	+	++	+		++	+	+	+	+
CPR	++	++	++	+	++	+		++	++	+	+	+
Dry mass	+	++	+		++			+		++		

variable, we defined a as 70% of the maximum recorded value and b as 80% of the maximum value of the medium outliers.

2.4.2. High frequency noise

In all measured variables, the high frequency component of the noise signal appears in clusters at different amplitudes and in different frequency bands. We developed a procedure to detect these clusters automatically and to estimate their characteristic parameters.

The MAD function (see Eq. (23)) was applied, in a moving window, to generate a signal that ''covers'' the noise curve, identifying a different cluster when noise exceeds this ''mapping'' signal. The procedure includes a tuneable threshold to identify every cluster in each measured signal. The noise of each cluster was then modelled as an autoregressive (AR) filter output driven by a white noise input, whose transfer function is

$$H(z) = \frac{\sqrt{e}}{1 + a_1 z^{-1} + a_2 z^{-2}} \quad (24)$$

where e represents the power of the white noise input, and a_1 and a_2 the characteristic coefficients of the AR filter. The Yule–Walker technique was used within the Matlab Signal Processing toolbox to estimate filter parameters (for details see [40]).

3. Results and discussion

Sensitivity analysis and calibration results of the deterministic model are presented first. Then we look at signal processing and noise model performance, and finally we compare real average bed temperature measurements with simulations obtained using the entire combined model.

Table 2
Fixed model parameters

Parameters	Values	Units
A	11.78	m^2
Cp_a	1004	J/kg K
Cp_{CO_2}	843.2	J/kg K
Cp_M	468	J/kg K
Cp_{O_2}	917.4	J/kg K
Cp_s	2000	J/kg K
Cp_v	1867	J/kg K
Cp_w	4187	J/kg K
g	9.8	m/s^2
k_{exp}	1.1	–
k_i	2.95E+005	$1/kg_{d.b}\ kg_{GA3}$
k_P	0.00169	1/h
L	2.5	m
L_w	2432000	J/kg_W
M_M	1000	kg
m_{O_2}	0.0554	$kg_{O_2}/kg_x\ h$
T_{max}	28	°C
T_{min}	18	°C
T_w	20	°C
T_{wall}	28	°C
w	12	–
Y_{x/CO_2}	0.6607	kg_x/kg_{CO_2}
$Y_{x/Ni}$	30	kg_x/kg_N
Y_{x/O_2}	0.9417	kg_x/kg_{O_2}
β_M	6.41E−004	$kg_{GA3}/h\ kg_x$
ε	0.22	–
ϕ	0.04349	–

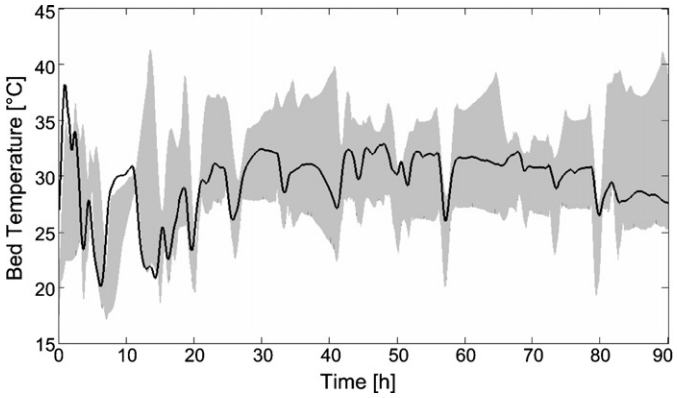


Fig. 4. Simulated average bed temperature (solid line) vs. range of measured bed temperatures (shadow).

Table 3
Fitting parameters

Parameters	Initial values	Optimal values	Units
δ	0.0214	0.022593	1/h
k	0.00192969	0.0020002	1/h
m_{CO_2}	0.1212002	0.10417	kg_{CO_2}/kg_x h
μ_M	0.20694318	0.21629	1/h
k_N	0.0012942	0.0012731	$1/kg_N\ kg_{d.b.}$
k_g	0.40	0.41664	$kg_{d,b}/kg_{CO_2}$
k_W	1.15	1.1529	kg_W/kg_{CO_2}
Y_{q/CO_2}	7023305.08	7038329.47	J/h
Fitting error	10427	10071	

3.1. Deterministic model

3.1.1. Sensitivity analysis

Table 1 summarizes the overall effect of parameter values on process variables. Model calibration parameters that have a significant effect on the values of the simulated variable, appear with a (++) sign, those with a (+) sign indicate a moderate influence while those that do not affect the respective variable are left blank. From the table, given their impact on multiple variables, we can see that the main model parameters are δ, k, k_N, Y_{q/CO_2}, m_{CO_2}, μ_M, and Y_{q/CO_2}. In addition, k_W and k_g are important model parameters, because they have a substantial effect on the simulation of two key process variables; bed water content and total dry mass. Parameters k_P, m_{O_2} and Y_{x/O_2} had little impact on simulation results, however.

3.1.2. Model calibration

Guided by the normalized errors defined in Eq. (21), initial estimates of model parameters were set by trial and error. For model optimal calibration eight parameters were selected based on the above analysis. The remaining model parameters (Table 2) were assigned fixed values from literature sources, process knowledge or preliminary calibrations.

The optimization routine took 5000 function evaluations to establish optimal parameter values; Table 3 displays a comparison between optimal values and their initial guesses. Figs. 4 and 5 illustrate the quality of the model fit.

Optimum parameters improved bed temperature fitting significantly, and ensured that simulations were always within the ranges of the measured values (Fig. 4). The model also reproduced the behaviour of the outlet air temperature well (Fig. 5d) and so endorses the heat transfer model employed. Fitting of total CO_2 production is also good (Fig. 5b). However, it was not possible to achieve a good fit for CPR; the peaks observed (Fig. 5c) are due to bed heterogeneity, which is overlooked in the simplified lumped parameter model used here. Another limitation is that bed water content is overestimated at the end of the fermentation (Fig. 5a). This could be curbed by assigning water content errors a higher cost function weighting, but the trade-off in doing so is deterioration in the bed temperature fit. Alternatively, a varying seepage term could be incorporated, but this would make the model overcomplicated.

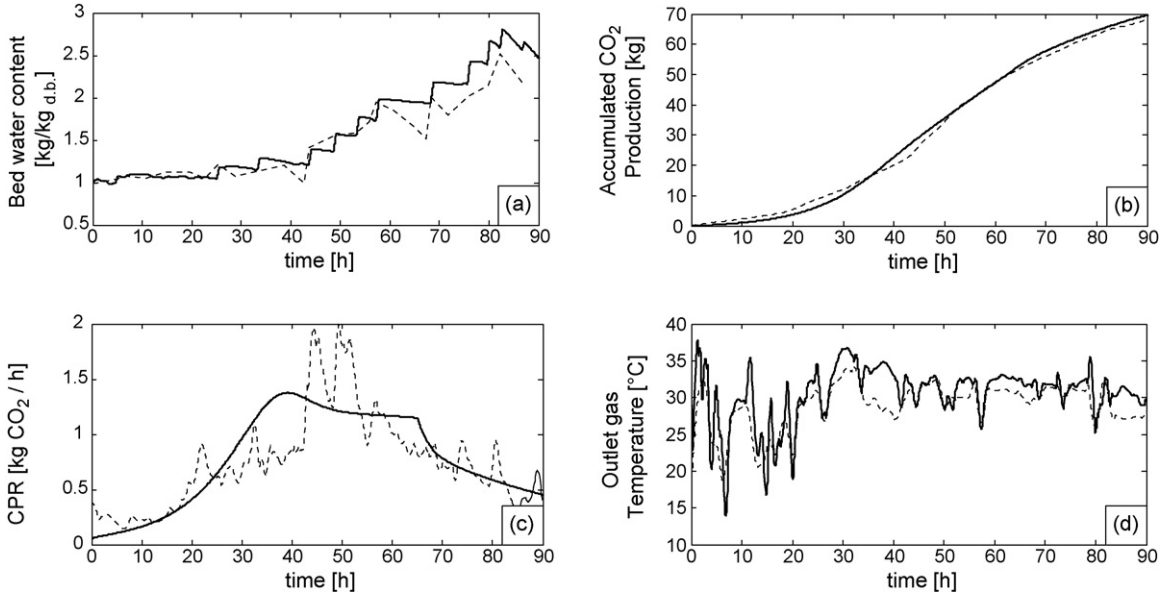


Fig. 5. Simulated (solid lines) vs. measured (dotted lines) values for bed water content (a), total CO_2 (b), CPR (c) and outlet air temperature (d).

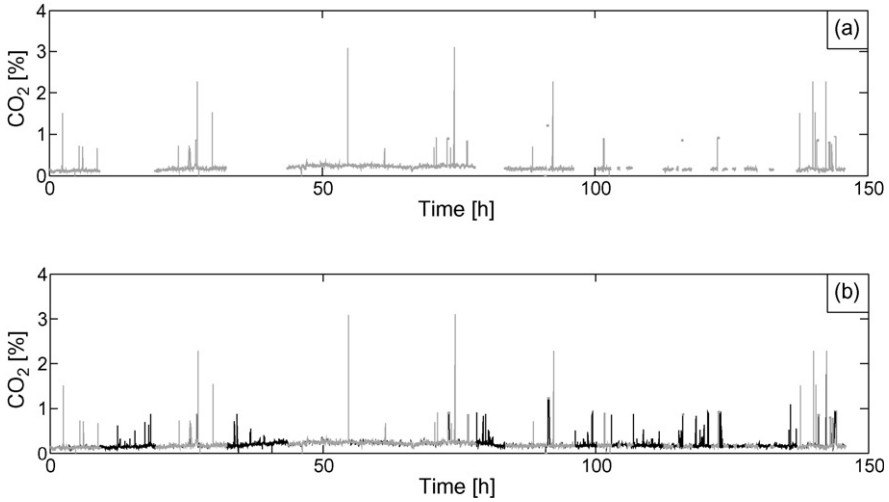


Fig. 6. Assignment of missing data: (a) original signal with missing data removed; (b) processed signal with assigned values (dark).

3.2. Signal processing

3.2.1. Missing data

Fig. 6a shows a typical signal (CO_2 concentration in this case) once missing values are removed. The processed signal after applying the missing data assignment algorithm described earlier is shown in Fig. 6b (assigned values are depicted in boldface). The trend of the assigned values appears similar to the measured values in the original signal. The proposed algorithm also provided a pattern of assigned outliers similar to that observed in the original signal. This complete form of reconstruction cannot be achieved with conventional interpolation techniques.

3.2.2. Outliers

Fig. 7 shows the outliers detected in a representative record of the inlet air temperature using window length as the outlier detection algorithm tuning parameter. If a large window length is used (around 8000 data points in Fig. 7a) only outliers with deviations larger than 10 °C were detected. Small windows (10 data points in Fig. 7c) returned many false outliers. With a window length of 100 we were able to detect all outliers without any false positives (Fig. 7b). Many other values were tried and we concluded that the window length had to be between 50 and 200 for inlet air temperature. Fig. 7c shows the resulting signal once outliers were removed using a window length of 100.

3.2.3. High frequency noise

For high frequency noise two tuning parameters are available; window length and polynomial order. We observed a relationship between both parameters. For example, we achieved good filtering with a small window length and a small order polynomial (1 and 51, respectively). However, we also obtained good filtering using a long window with a high order polynomial (3 and 101, respectively); both filtered signals looked practically identical (not shown).

3.3. Noise modelling

3.3.1. Outliers

Fig. 8 compares outliers detected by the Hampel filter [37] with outliers generated by the model for a historical record of inlet air temperature. For this particular case, the outlier model's parameters were: four positive high level outliers at 4390 °C and 27 medium level outliers at 370 °C. Low level outliers were generated randomly using a Gaussian distribution (amplitude) and an exponential distribution (time). Although the distribution of outliers in Fig. 8 seems different, they are similar from a statistical point of view. This becomes more apparent in Fig. 9 where the statistics for the low level outlier distribution of the real signal (Fig. 9a and b) are compared with that of the simulated signal (Fig. 9c and d). The figure shows the measured (histogram) and estimated (solid line) distribution of real and simulated signals. We see that the estimated amplitude

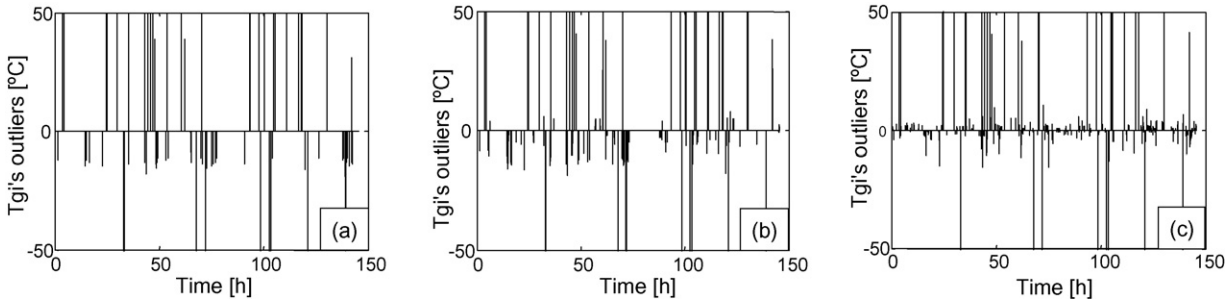


Fig. 7. Outliers detected by Hampel filter using different window lengths (high outliers are not shown). (a) Maximum length; (b) 100 points; (c) 10 points.

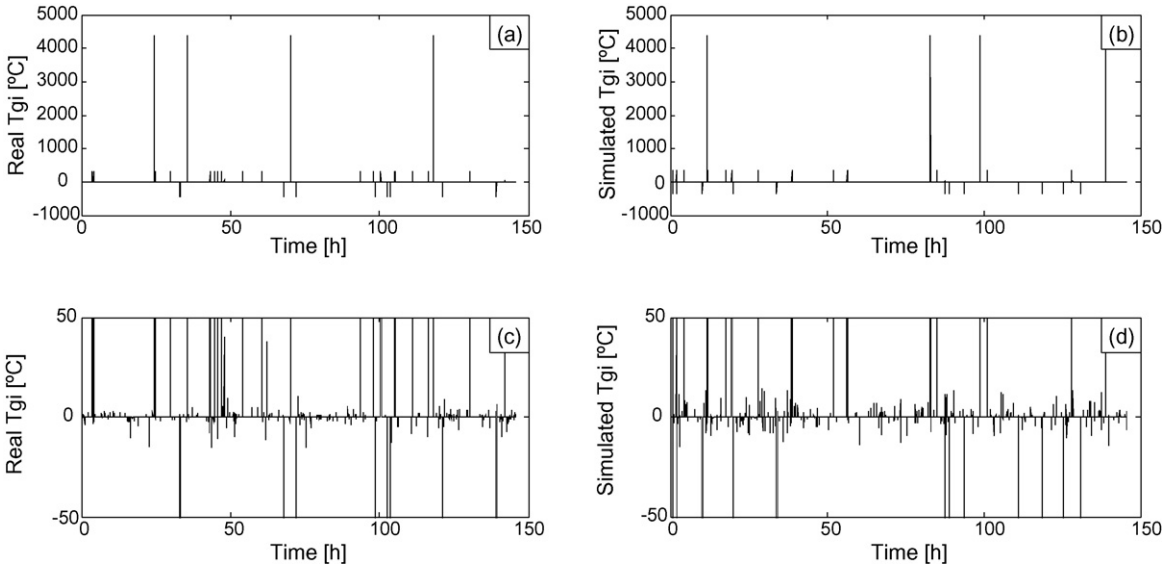


Fig. 8. Real outliers (left) vs. model generated outliers (right). Lower signals show small outliers more clearly.

and time distributions of both signals are practically identical and, as such, the low level outlier model represents the real signal well.

3.3.2. High frequency noise

Modelling the high frequency noise involved many steps (Fig. 10). Fig. 10a illustrates the high frequency noise contaminating the inlet air temperature of a given fermentation, and clearly shows the high and low amplitude noise clusters. A modified MAD algorithm with a tuneable threshold described earlier was applied (Fig. 10b) and automatically detected 17 noise clusters (Fig. 10c). Different threshold values will produce different cluster sets so the user needs to choose given the trade-off between the number of clusters and model complexity. Finally, Fig. 10d shows that the simulated noise signal the outlined high frequency noise model generated is similar to the real noise produced (Fig. 10a).

3.4. Realistic model simulation

The average bed temperature is the most important and difficult variable to control in an SSF bioreactor. We expect our simulations to reproduce the observed dynamics of this variable so as to permit the design of effective temperature control strategies. Figs. 11 and 12 compare real average bed

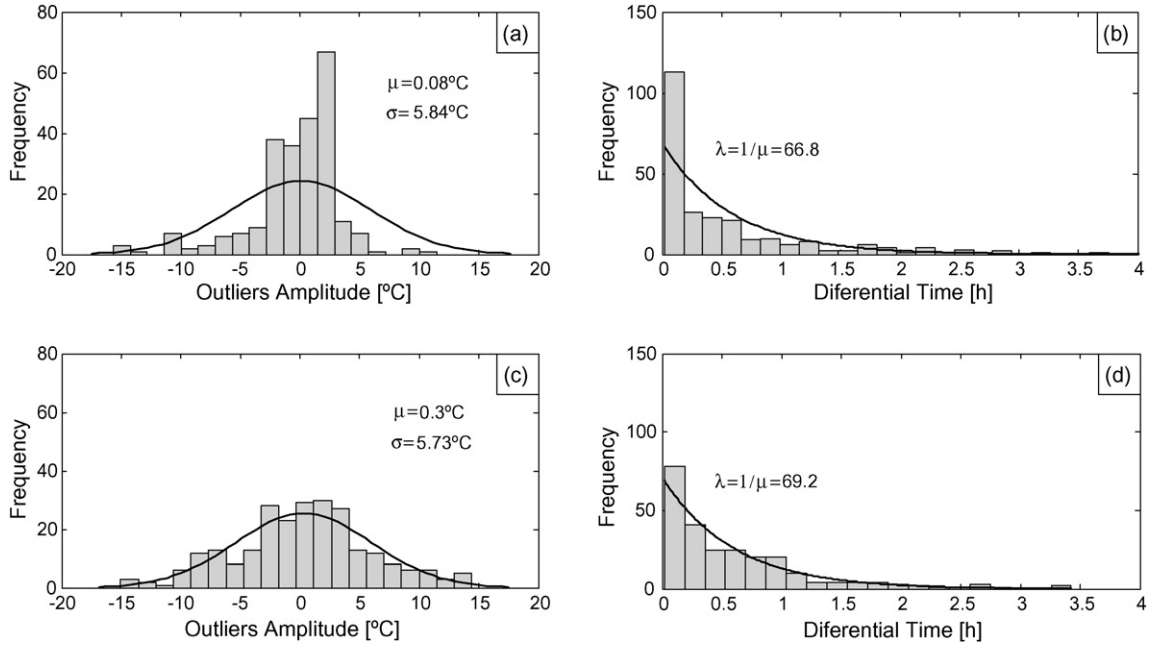


Fig. 9. Probability distribution functions (PDFs) of low level outliers. (a) PDF of real outlier amplitudes; (b) PDF of real outlier occurrences over time; (c) PDF of simulated outlier amplitudes; (d) PDF of simulated outlier occurrences over time.

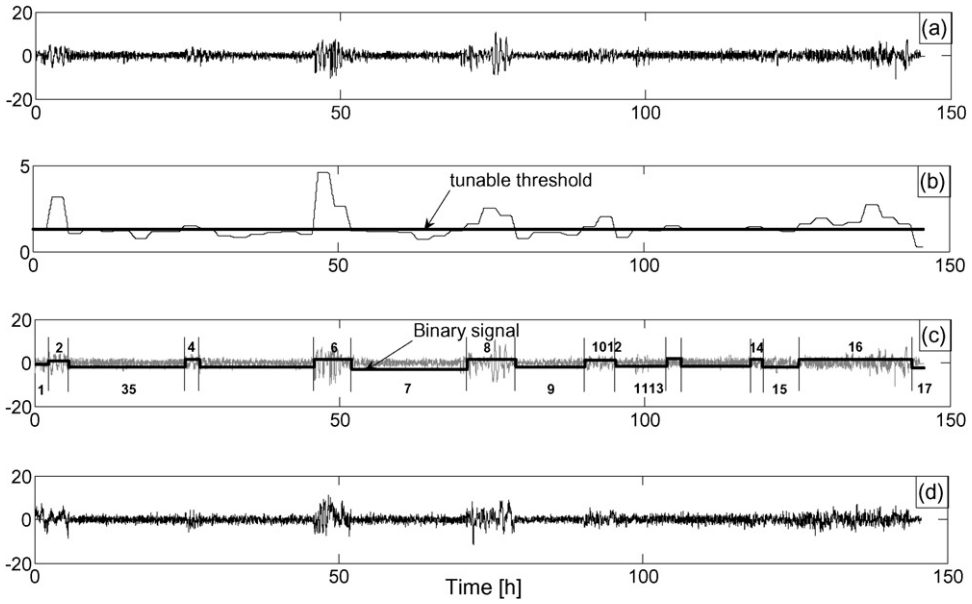


Fig. 10. (a) True noise; (b) wrapping signal for a given threshold (window length = 150); (c) 17 noise clusters found marked by a binary signal; (d) simulated noise.

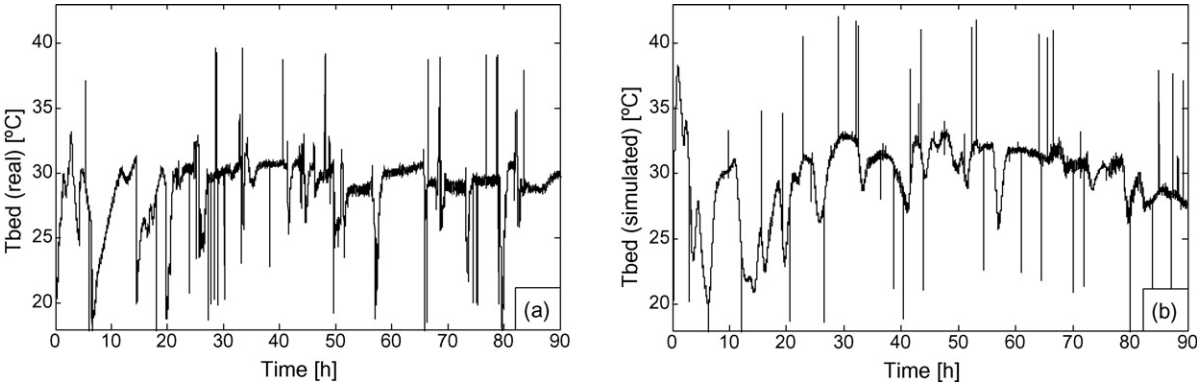


Fig. 11. Real (a) vs. estimated (b) bed temperature during the whole fermentation showing the general trend and the outliers.

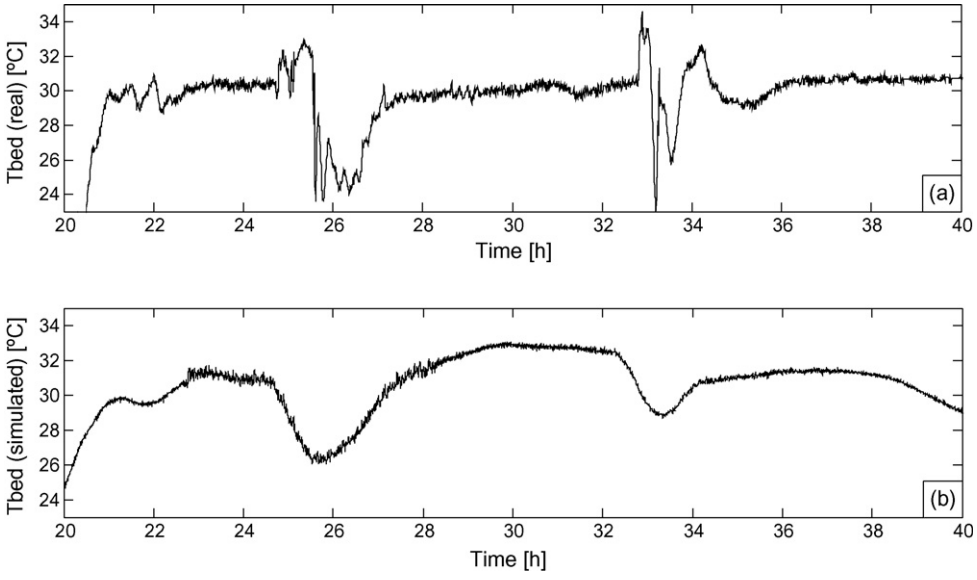


Fig. 12. Real (a) vs. estimated (b) bed temperature during 20 h of fermentation showing high frequency noise (outliers removed).

temperature data (including assigned missing data) with simulations drawn from the "realistic" (phenomenological plus noise) model. Both signals, real (Fig. 11a) and simulated (Fig. 11b), present the same dynamic trend and similar outlier distribution, in terms of quantity and amplitude. For average bed temperature we normally encounter far fewer outliers than in other fermentation variables; even so the outlier model does represent the real data well. Showing just few hours of the fermentation, Fig. 12 focuses on high frequency noise model behaviour with outliers removed. Both signals returned the same number of high frequency noise clusters of similar amplitude. Fig. 12a, nevertheless, shows a medium frequency signal noise that is not observed in the simulated signal (Fig. 12b). This noise is due to bed heterogeneity and bed mixing events, which cannot be reproduced with a lumped parameter model.

3.5. Concluding remarks

This is the first time a combined, realistic phenomenological-noise model fitted with experimental data has been proposed for solid-state fermentation bioreactors. Incorporating the realistic noise model makes it possible to attain a usefully close representation of the complex behaviour observed in real measurements from a pilot scale SSF bioreactor. Previous models of SSF bioreactors, such as in [41], consider overly simplistic stationary noise, i.e., constant amplitude and frequency band, as well as lacking outliers. In our experience at controlling 50 and 200 kg SSF pilot bioreactors, which is supported by the extensive literature on noise-control interactions (see for example [42]), the design of control systems based on process simulations using simplistic noise models results in poor real-process control performance. Incorporating a complex noise model (better able to represent the observed or expected noise pattern) into the phenomenological bioreactor model produced realistic simulations. These, in turn, should prove to be useful tools when designing and tailoring control systems for a specific SSF process, as they should ensure more effective and reproducible plant operation.

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